SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

CHENNAI – 602105

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – INDUSTRY PROBLEM SOLVING

Course Code:	DSA0502	**Course Title:**	Query Processing for Data Science
CO Assessed:	CO5	**Assessment:**	Tool 2– **INDUSTRY PROBLEM SOLVING**
Weightage:	35%	**Total Marks:**	

CO5 Statement: Create meaningful visualizations using Pandas and Matplotlib, including charts, distributions, relationships, and interactive visual representations for effective communication

Student Name:	Sandra Sudevan	**Reg. No.:**	192424422
Section / Batch:	2024	**Date of Submission:**	24/08/26
Faculty In-charge:	**Dr. B.Shyamala Gowri**	**Project Mode:**	Individual Submission

Code of Conduct

I Sandra Sudevan(192424422)(Name / Reg No) certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature: Sandra Sudevan

1. **Manufacturing Production Analysis**
   A manufacturing company records **daily production, machine operating hours, temperature, energy consumption, and defect count**. Import and explore the dataset using Pandas. Develop visualizations to analyze production trends over

time, relationships between machine parameters and defects, and distributions of production values. Use a **scatter matrix, lag plot, and autocorrelation plot** to identify hidden production patterns and provide recommendations for improving efficiency.

The dataset contains information such as **date, production quantity, machine operating hours, temperature, energy consumption, and defect count**.

1. **Import and explore the dataset**
   Pandas is used to load and examine the data. Basic information such as the number of records, column names, data types, and statistical values is checked.
2. **Analyze production trends over time**
   A **line chart** is used to observe how production changes from day to day. This helps identify increasing, decreasing, or unusual production patterns.
3. **Study machine parameters and defects**
   Scatter plots are used to examine relationships between variables such as machine operating hours, temperature, energy consumption, and defect count. This helps identify conditions that may increase defects.
4. **Analyze production distribution**
   A **histogram** shows how production values are distributed. It helps determine whether most days have similar production levels or whether there are unusually high or low values.
5. **Use a scatter matrix**
   A **scatter matrix** compares multiple numerical variables at the same time. It helps identify relationships between production, machine hours, temperature, energy consumption, and defects.
6. **Use a lag plot**
   A **lag plot** compares today's production with the previous day's production. If the points show a clear pattern, it indicates that production may depend on previous-day production.
7. **Use an autocorrelation plot**
   The **autocorrelation plot** measures how strongly production values are related to their previous values at different time lags. It helps detect repeating or time-dependent production patterns.
8. **Provide recommendations**
   Based on the visualizations, the company can identify factors affecting production and defects. For example, controlling machine operating conditions, monitoring energy consumption, and maintaining machines regularly can help **reduce defects and improve production efficiency**.

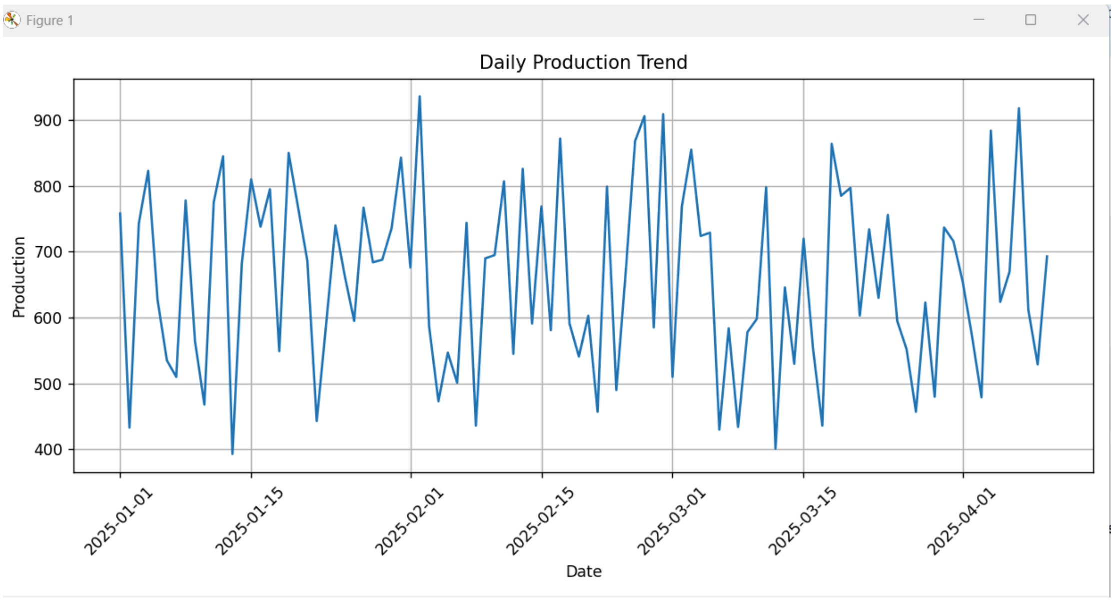


Figure 1
Daily Production Trend
Production
Date
900
800
700
600
500
400
2025-01-01
2025-01-15
2025-02-01
2025-02-15
2025-03-01
2025-03-15
2025-04-01

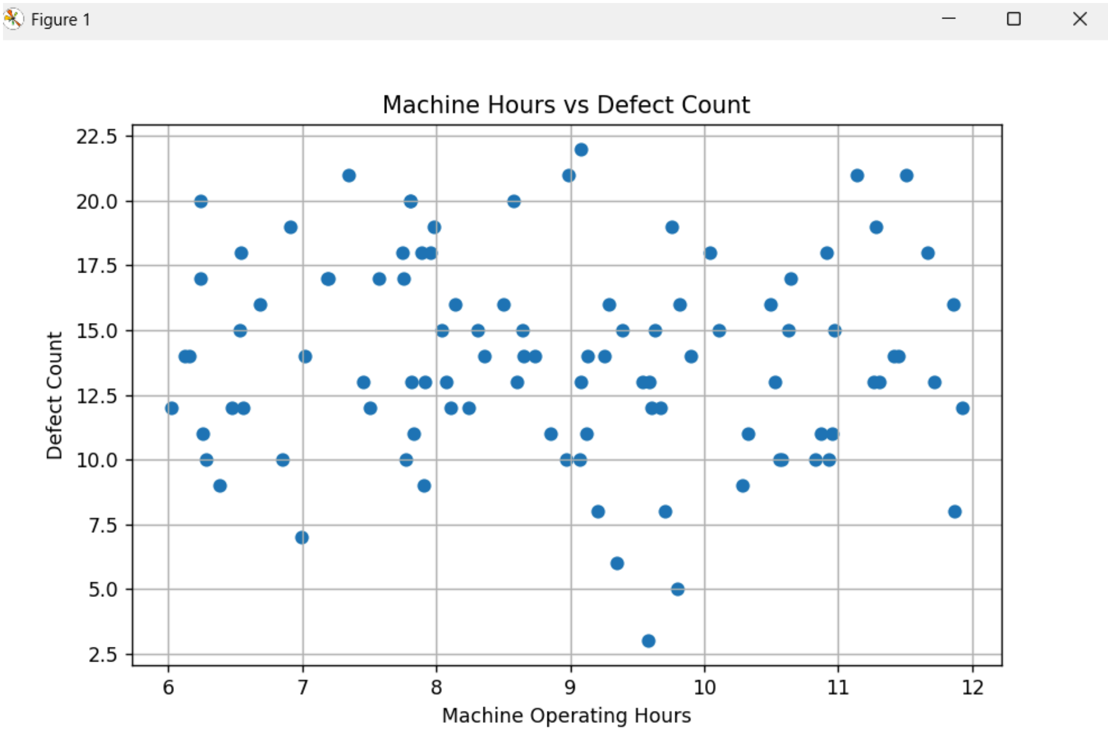


Figure 1
Machine Hours vs Defect Count
Defect Count
Machine Operating Hours
22.5
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
6
7
8
9
10
11
12

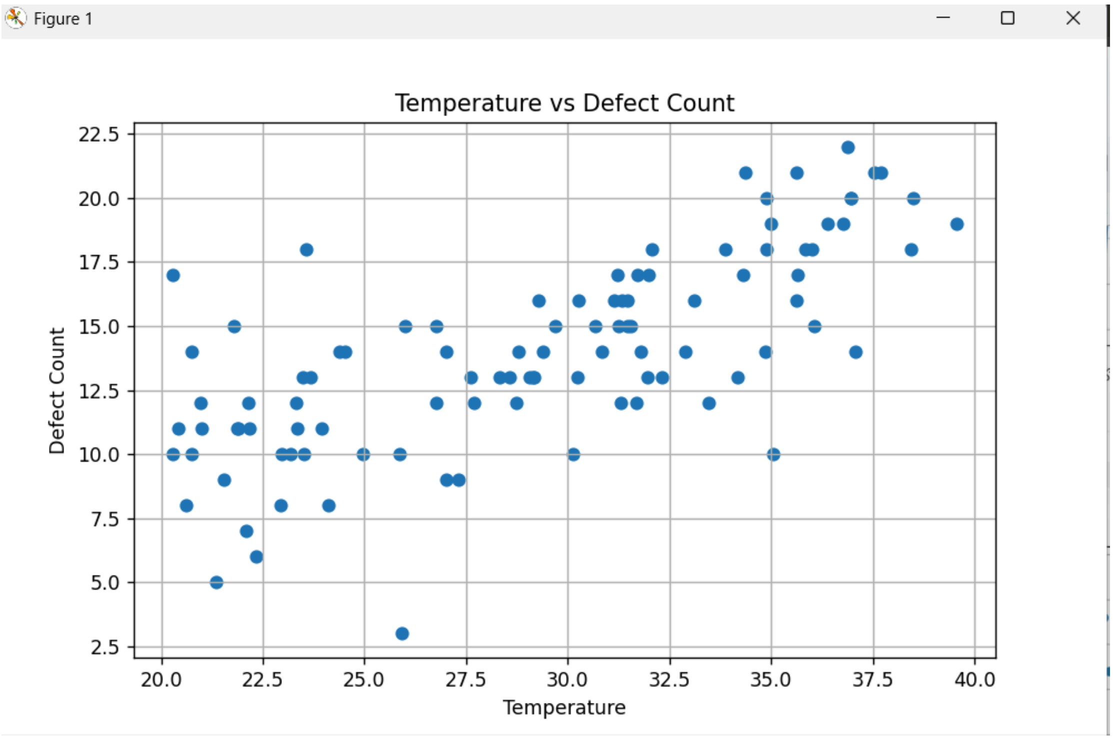


Figure 1
Temperature vs Defect Count
Defect Count
Temperature
22.5
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
20.0
22.5
25.0
27.5
30.0
32.5
35.0
37.5
40.0

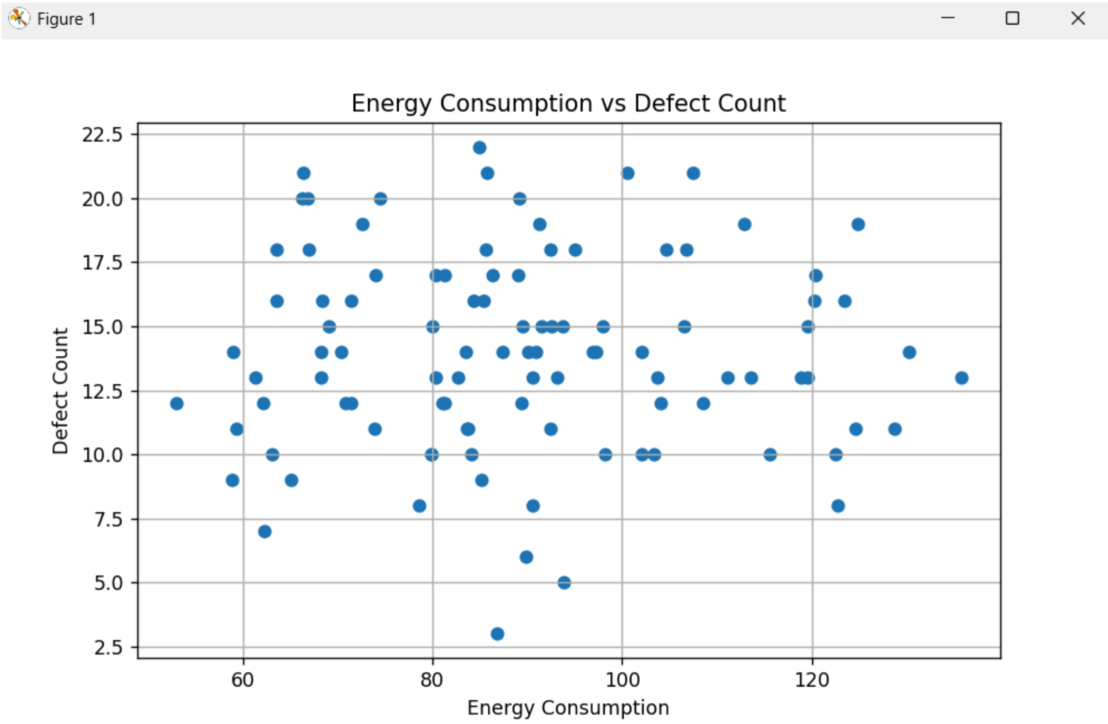


Figure 1
Energy Consumption vs Defect Count
Defect Count
Energy Consumption
22.5
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
60
80
100
120

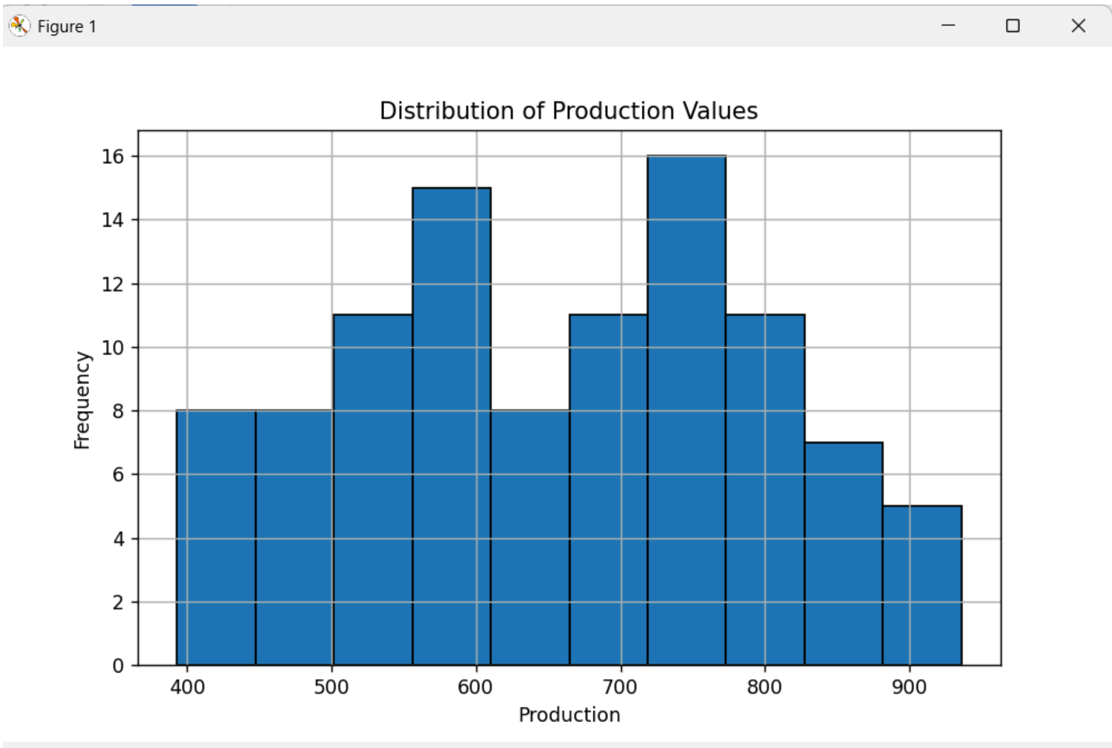


Figure 1
Distribution of Production Values
Frequency
Production
0
2
4
6
8
10
12
14
16
400
500
600
700
800
900

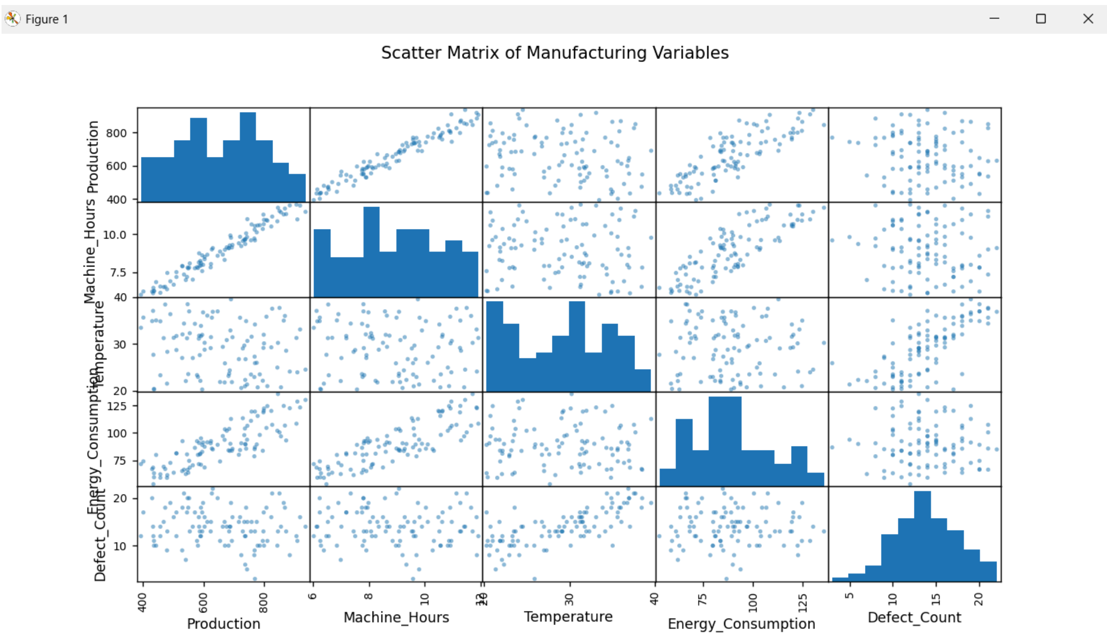


Figure 1
Scatter Matrix of Manufacturing Variables
Production
Machine_Hours
Temperature
Energy_Consumption
Defect_Count

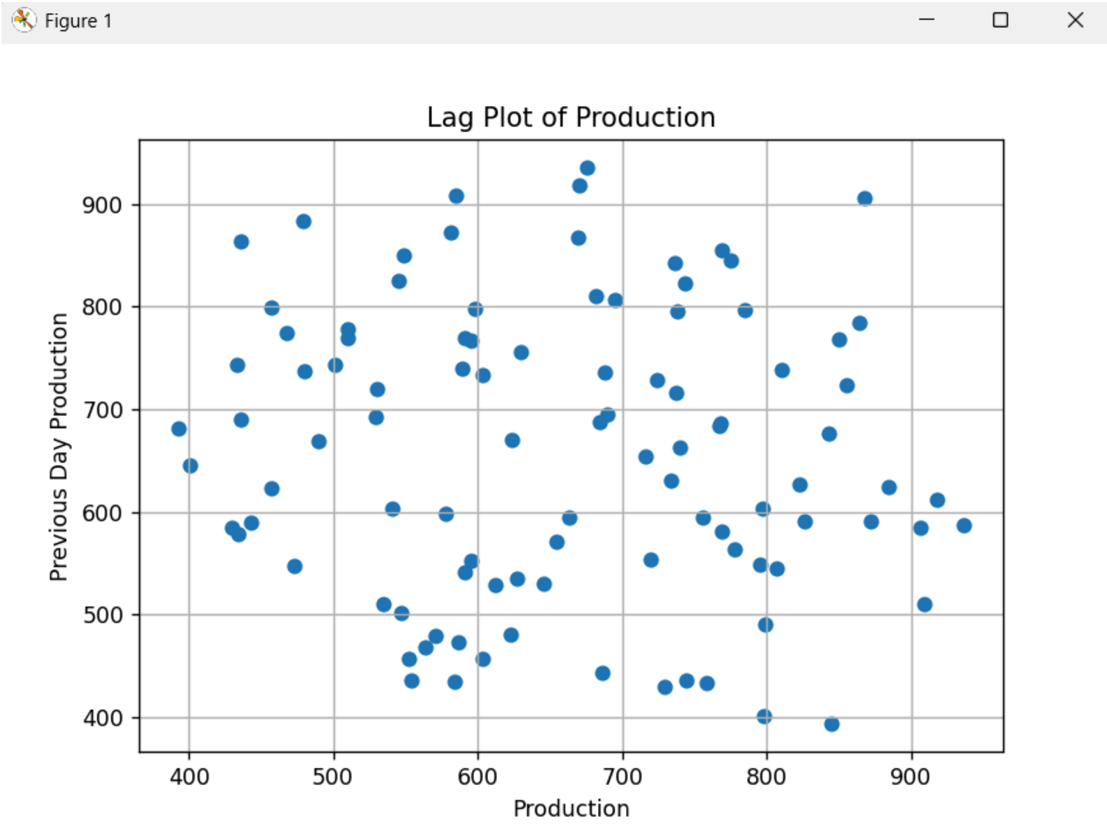


Figure 1
Lag Plot of Production
Previous Day Production
900
800
700
600
500
400
400
500
600
700
800
900
Production

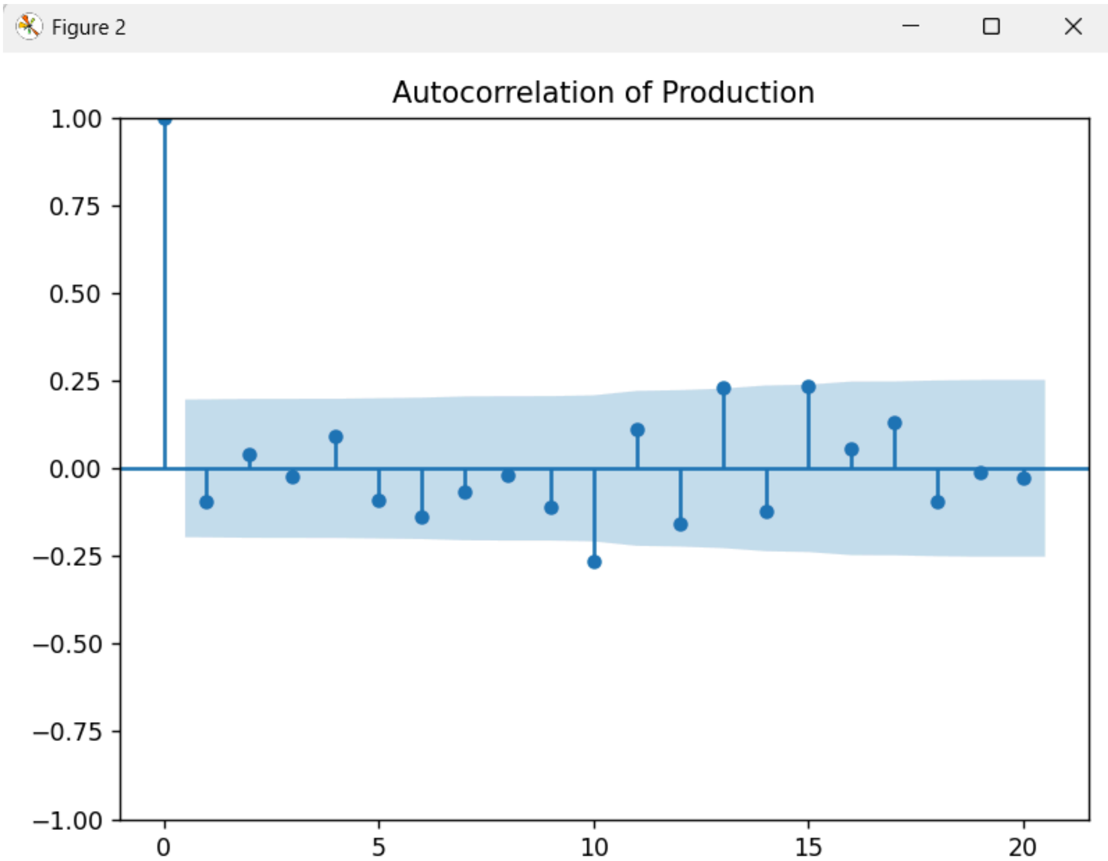


Figure 2
Autocorrelation of Production
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
5
10
15
20

Overall: The analysis combines Pandas and visualization techniques to understand production performance, detect hidden patterns, identify possible causes of defects, and support better manufacturing decisions.

2. **E-Commerce Sales Pattern Analysis**
   An e-commerce company has historical data containing **date, product category, quantity sold, revenue, discount, and customer rating**. Analyze the evolution of sales over time and identify relationships between discount, quantity, and revenue. Use appropriate **bar charts, line charts, distribution plots, frequency plots, and scatter matrices** to identify important business patterns.

The dataset contains information such as **date, product category, quantity sold, revenue, discount, and customer rating**.

1. **Import and explore the dataset**
   Pandas is used to load the e-commerce data and understand its structure. The first few records, data types, missing values, and statistical summary are examined.
2. **Analyze sales over time**
   A **line chart** is used to show how sales or revenue changes over different dates. This helps identify increasing or decreasing sales trends.
3. **Compare product categories**
   A **bar chart** is used to compare sales or revenue among different product categories. This helps identify the best-performing categories.
4. **Analyze data distributions**
   Distribution plots and histograms are used for variables such as quantity sold, revenue, discount, and customer rating. They show how frequently different values occur.
5. **Study discount and quantity relationships**
   Scatter plots are used to examine whether discounts influence the quantity of products sold.
6. **Study discount and revenue relationships**
   The relationship between discount and revenue is analyzed using scatter plots. This helps determine whether higher discounts are associated with higher or lower revenue.
7. **Use a scatter matrix**
   A **scatter matrix** compares several numerical variables together, such as quantity, revenue, discount, and customer rating. It helps identify important relationships and patterns.
8. **Identify business patterns**
   The visualizations help determine which products sell more, how discounts affect sales, which periods have higher revenue, and whether customer ratings are related to sales.
9. **Provide recommendations**
   Based on the analysis, the company can improve pricing and discount strategies, focus on high-performing categories, and identify periods with high sales demand.

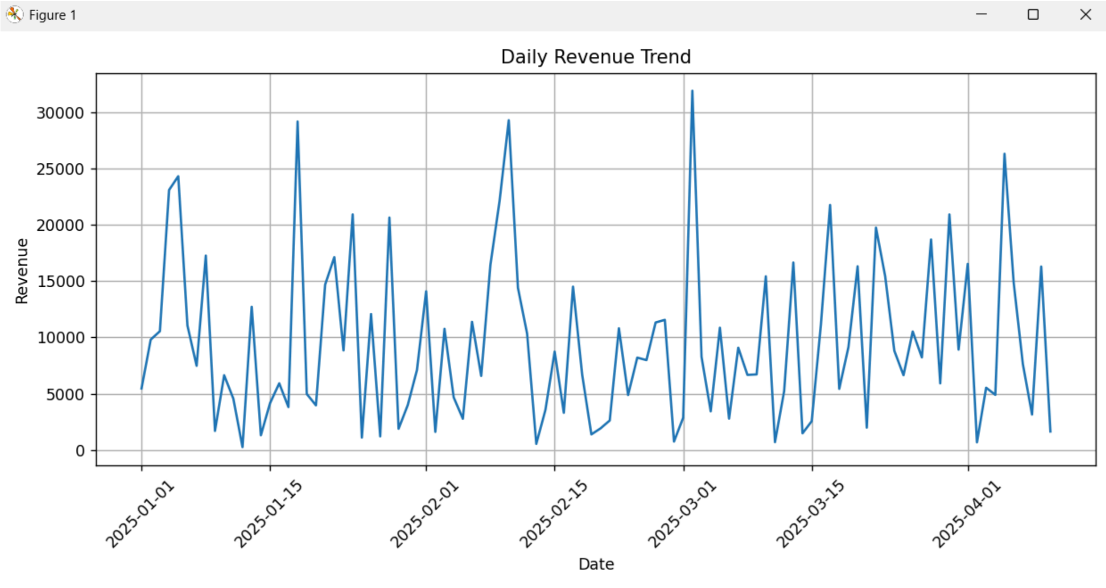


Figure 1
Daily Revenue Trend
Revenue
Date
0
5000
10000
15000
20000
25000
30000
2025-01-01
2025-01-15
2025-02-01
2025-02-15
2025-03-01
2025-03-15
2025-04-01

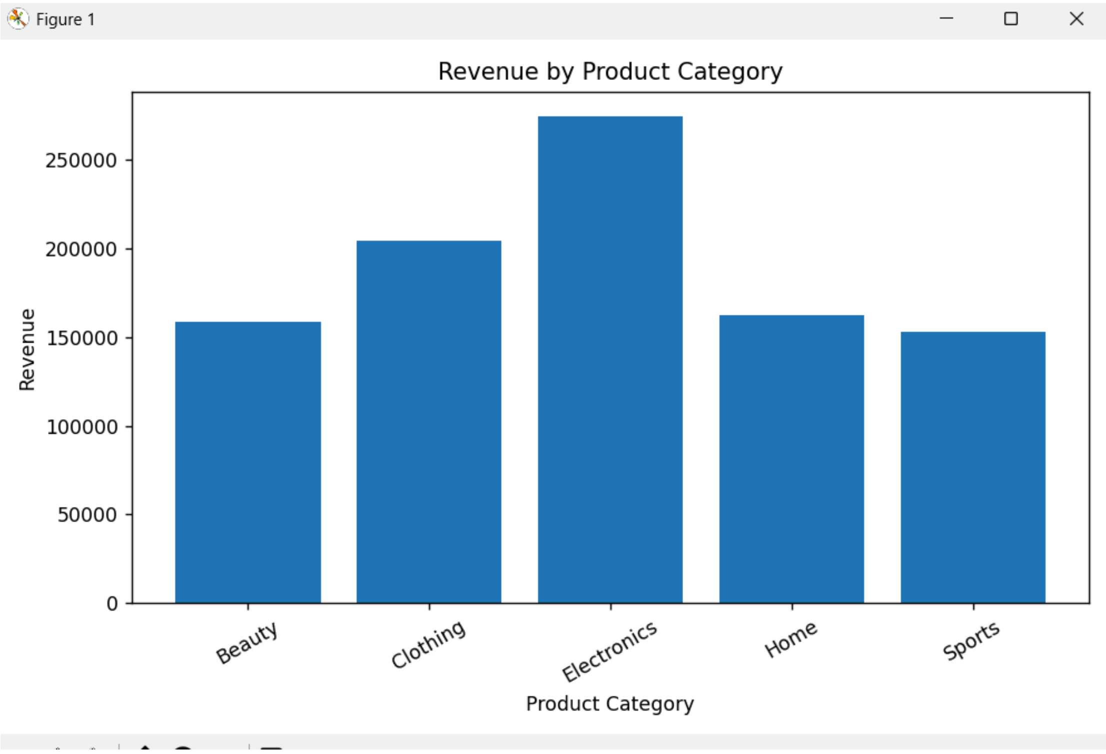


Figure 1
Revenue by Product Category
Revenue
Product Category
0
50000
100000
150000
200000
250000
Beauty
Clothing
Electronics
Home
Sports

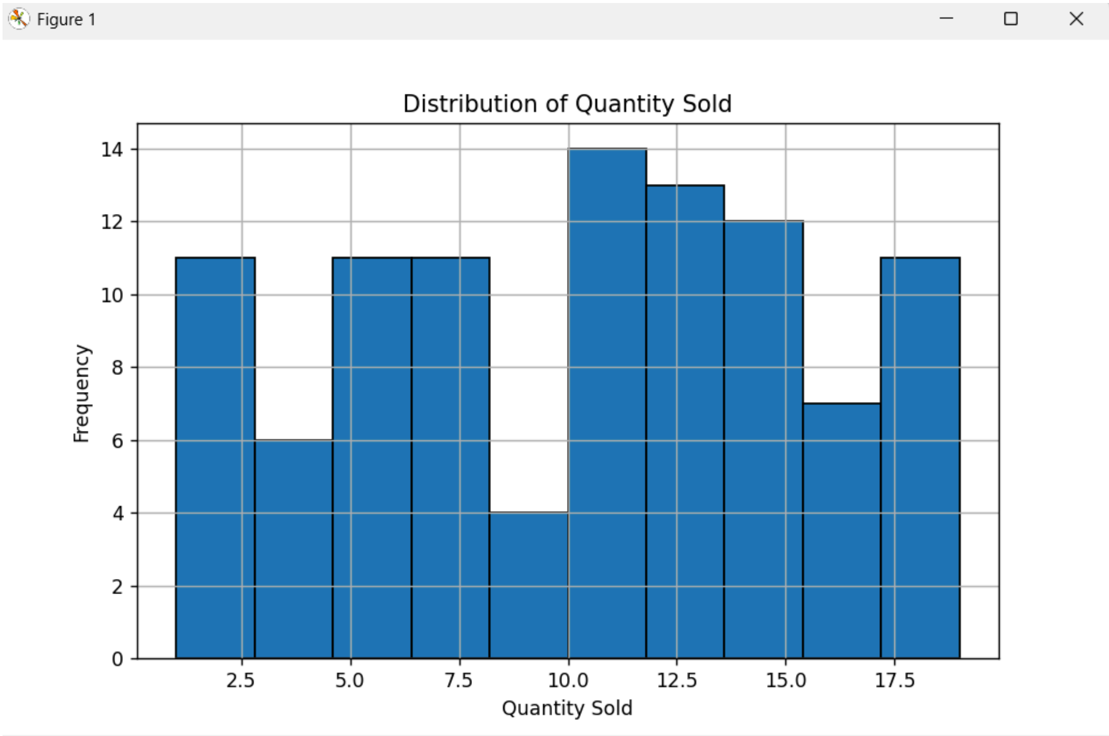


Figure 1
Distribution of Quantity Sold
Frequency
0
2
4
6
8
10
12
14
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Quantity Sold

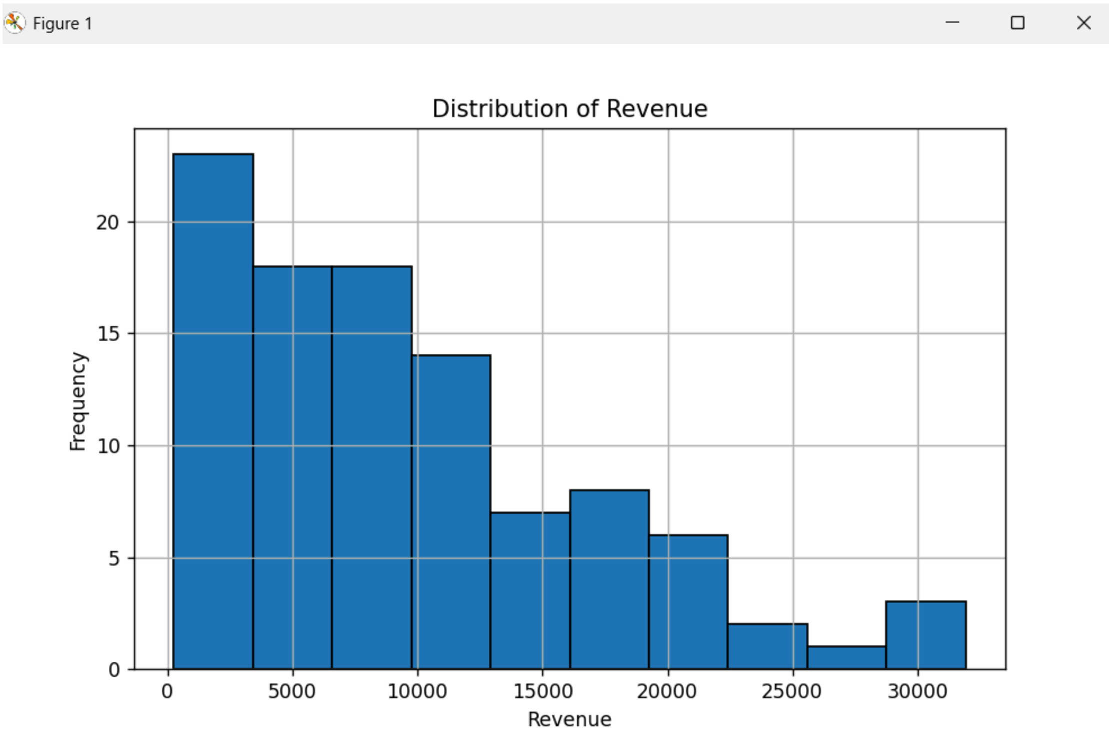


Figure 1
Distribution of Revenue
Frequency
0
5
10
15
20
0
5000
10000
15000
20000
25000
30000
Revenue

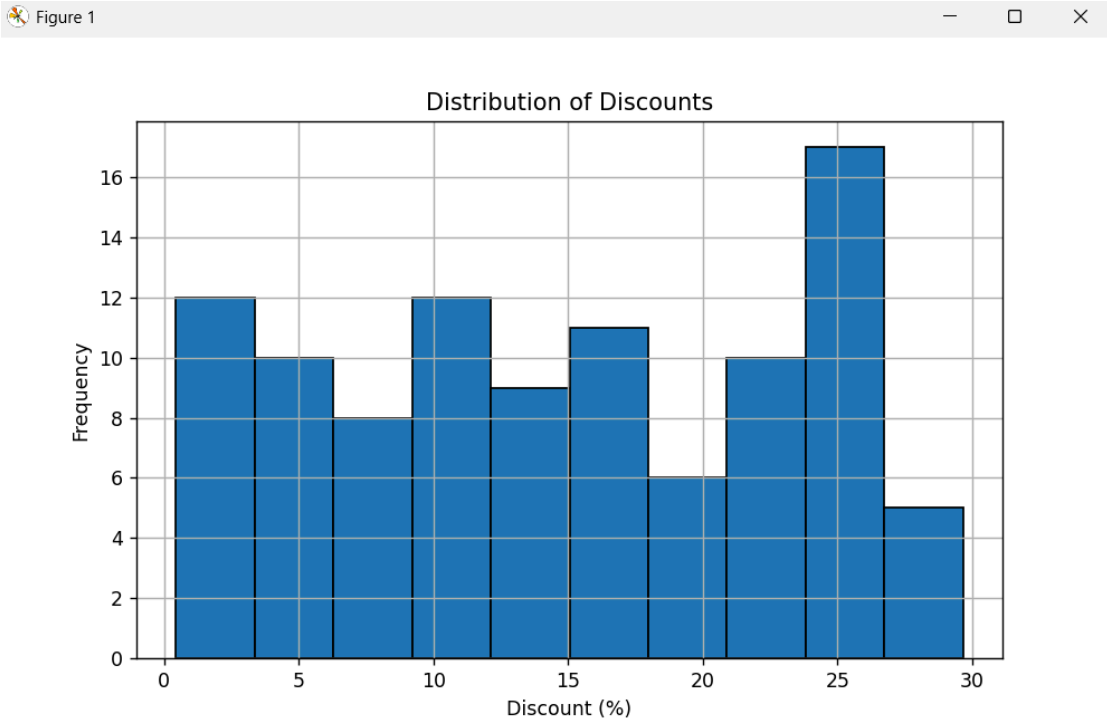


Figure 1
Distribution of Discounts
Frequency
Discount (%)
0
2
4
6
8
10
12
14
16
0
5
10
15
20
25
30

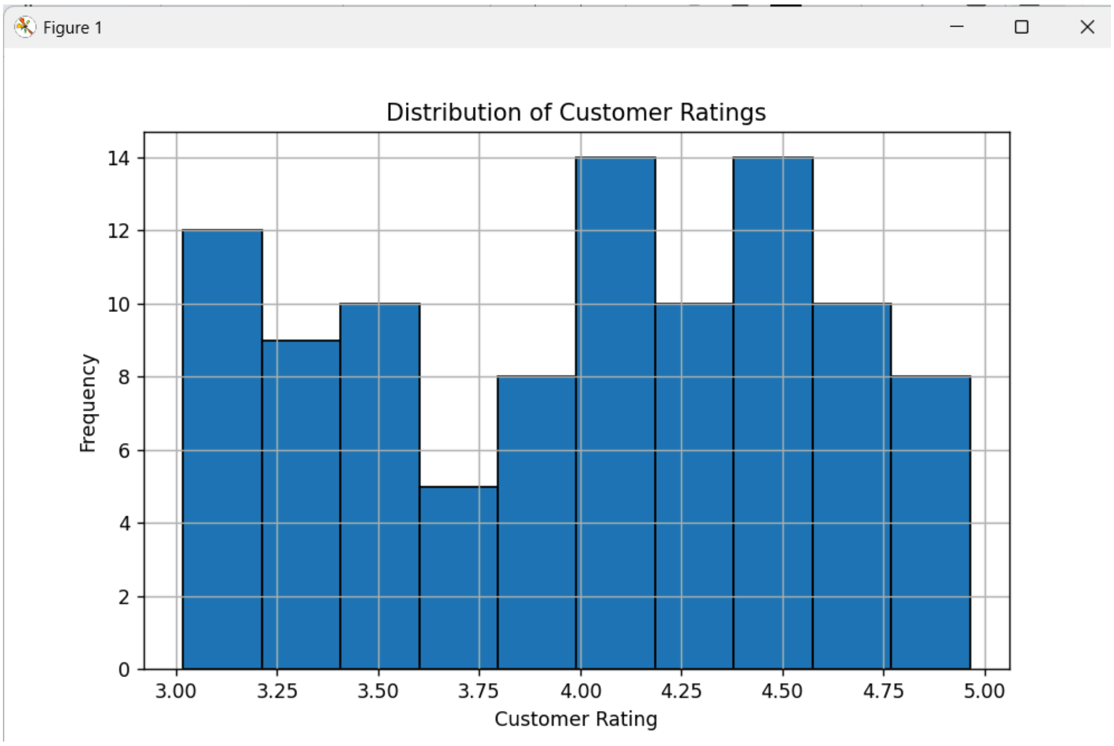


Figure 1
Distribution of Customer Ratings
Frequency
Customer Rating
0
2
4
6
8
10
12
14
3.00
3.25
3.50
3.75
4.00
4.25
4.50
4.75
5.00

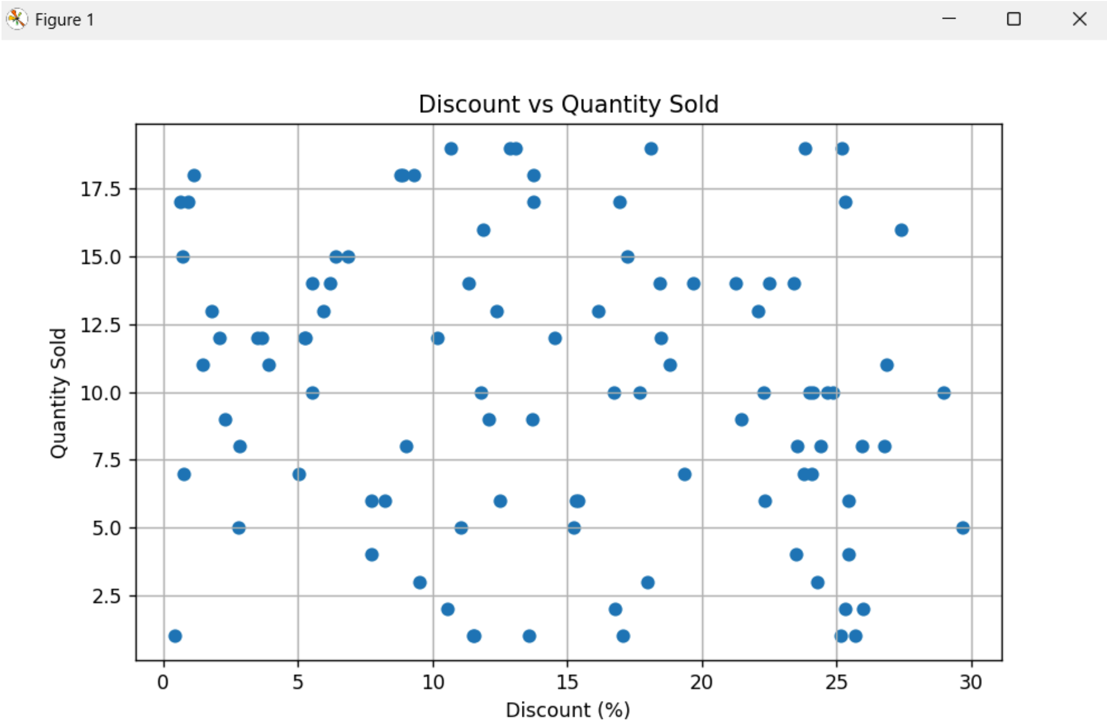


Figure 1
Discount vs Quantity Sold
Quantity Sold
Discount (%)

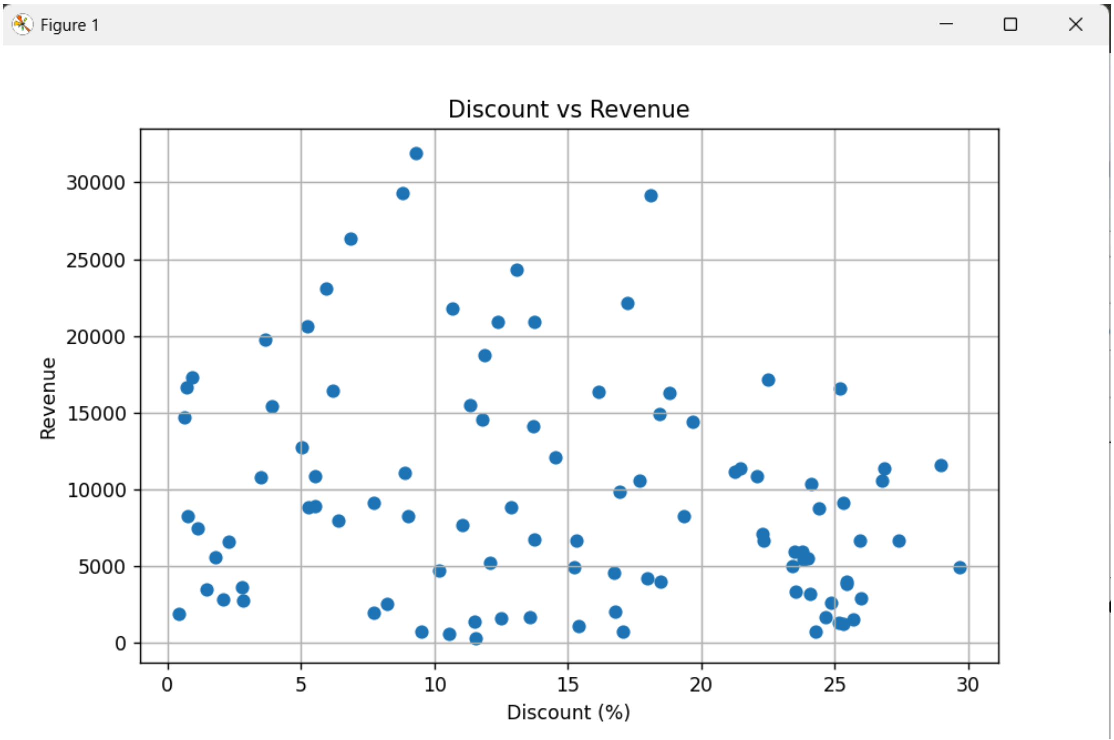
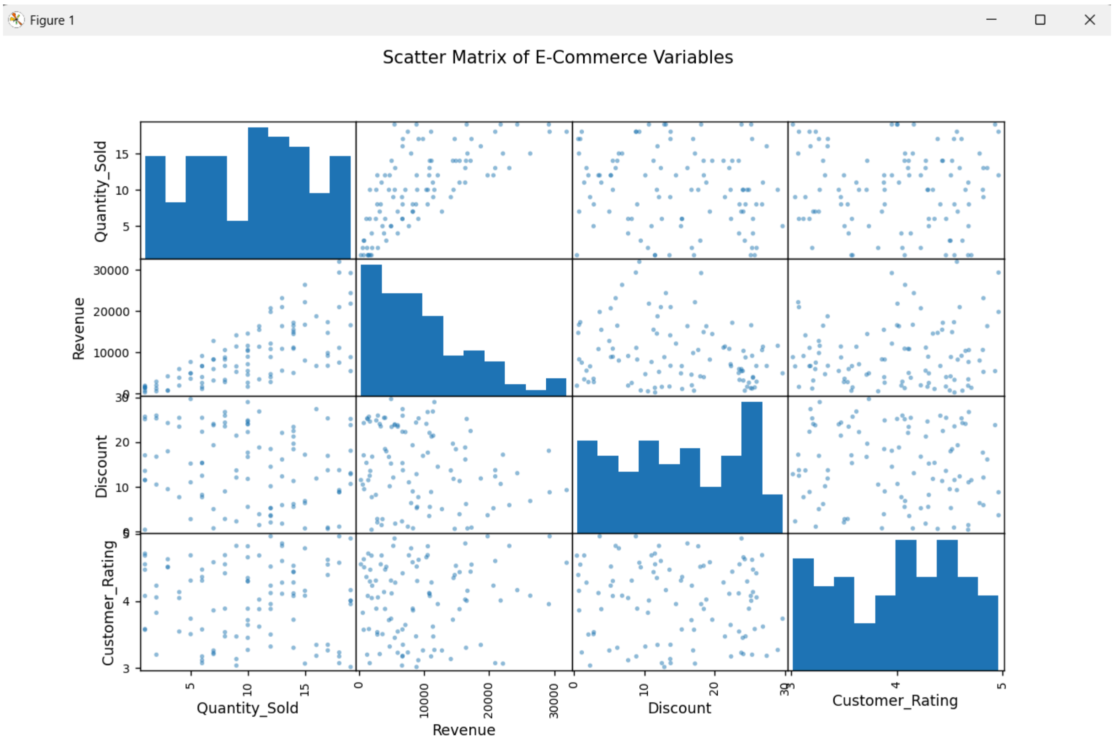


Figure 1
Discount vs Revenue
Revenue
Discount (%)



Overall: The analysis helps the e-commerce company understand **sales trends, customer behavior, product performance, and the effect of discounts**, which can be used to make better business decisions.